Lecture 41

Flux, surfaces and vector derivatives

Uses.

Summary 8: Scalar product, trigonometric and chain differentiation rules hold.
Summary 13: The cross product is perpendicular to its two factors, has parallelogram-area norm and reverses sign when the factors are exchanged.
Summary 26: Proper continuous integrals obey substitution and the fundamental theorem.
Summary 36: Gradients and Jacobians describe scalar and vector derivatives; the matrix chain rule holds.
Summary 37: Mixed second partial derivatives agree for a C^2 function.
Summary 38: M gives proper compact integration and admissible planar coordinate changes.
Summary 39: Spherical coordinates use θ from the positive z axis and φ as azimuth; compact balls have volume $4\pi R^3/3$.
Summary 40: Positive planar boundary orientation keeps the region on the left; Green equates circulation to the scalar-curl area integral on its stated domains.

Objectives. *Green's theorem computes area and planar outward flux from a boundary. For a surface in space, flux needs both an area factor and a choice of normal direction. We define these quantities through parametrisations, check how coordinate changes affect them, and compute a sphere and a graph. The Cartesian differential operators prepare the boundary laws of Lecture 42.*

By the end of this lecture you will be able to:

- Derive area and planar flux formulas from Green, and distinguish the two directions of a phase-plane orbit.
- Compute an oriented surface element and explain its behaviour under admissible parameter changes.
- Evaluate scalar surface integrals and vector flux on spheres and graph patches with the chosen normal.
- Compute divergence, curl and the Cartesian Laplacian, and prove the two Schwarz identities under C^2 hypotheses.

41.1 Area and a phase-plane integral

Let D satisfy the proved graph-region hypotheses of Green in Lecture 40, with positive boundary orientation. Apply it first to $(P, Q) = (0, x)$, then to $(-y, 0)$. Both scalar curls

equal 1, giving

$$\operatorname{area}(D) = \oint_{\partial D} x\,\mathrm{d}y = -\oint_{\partial D} y\,\mathrm{d}x = \frac{1}{2}\oint_{\partial D} (x\,\mathrm{d}y - y\,\mathrm{d}x).$$

These are proved formulas for those regions [P]. If the domain needs the finite-union or holes extension of Green, this route inherits [S]. Reversing orientation reverses every displayed boundary integral.

In a phase plane, the horizontal coordinate is q and the vertical coordinate is p. The same formula says

$$\oint_{\partial D} p\,\mathrm{d}q = -\operatorname{area}(D) \quad \text{for anticlockwise traversal in the } (q,p) \text{ plane.}$$

The order $p\,\mathrm{d}q$ fixes this sign. For example, let $m_0, \omega, E > 0$ and consider the oscillator energy ellipse

$$\frac{p^2}{2m_0} + \frac{m_0\omega^2 q^2}{2} = E.$$

Its horizontal semiaxis is $A = \sqrt{2E/(m_0\omega^2)}$ and vertical semiaxis is $m_0\omega A$. A linear change from the unit disc gives its area as $\pi m_0 \omega A^2 = 2\pi E/\omega$. Thus anticlockwise traversal gives $\oint p\,\mathrm{d}q = -2\pi E/\omega$. The actual oscillator parametrisation

$$q(t) = A\cos(\omega t), \qquad p(t) = -m_0\omega A\sin(\omega t), \qquad 0 \le t \le 2\pi/\omega$$

starts at the rightmost point and moves downwards: it is clockwise. Directly, $p\dot{q} = m_0\omega^2 A^2 \sin^2(\omega t)$, hence its integral is $\pi m_0\omega A^2 = 2\pi E/\omega$. Figure 41.1 displays the direction of this parametrisation.

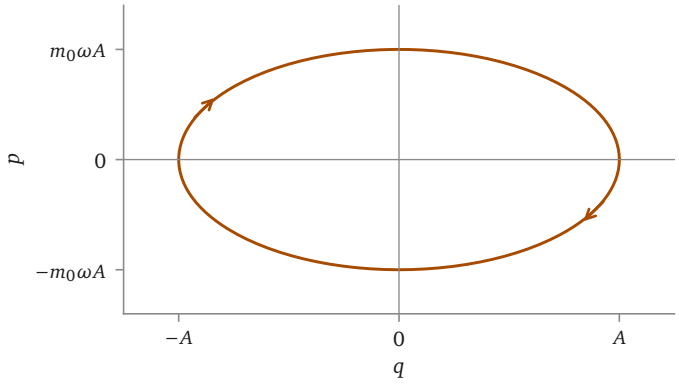


Figure 41.1: The oscillator trajectory starts at $(A, 0)$ and moves clockwise in the (q, p) plane. The integral $\oint p\,\mathrm{d}q$ for this traversal is the positive enclosed area.

The positive physical value here is consistent with the negative sign for the opposite, anticlockwise orientation.

41.2 Planar outward flux

For a regular positively oriented boundary piece $\gamma(t) = (x(t), y(t))$, the unit tangent points along (x', y'). The region lies to its left, so the outward unit normal points right:

$$\boldsymbol{n} = \frac{(y', -x')}{\sqrt{(x')^2 + (y')^2}}, \qquad \boldsymbol{n}\,\mathrm{d}s = (\mathrm{d}y, -\mathrm{d}x).$$

Here $\mathrm{d}s = \|\gamma'\|\,\mathrm{d}t$ is the scalar length element. Values at the finitely many corners do not change the integral. The *outward flux* of $\boldsymbol{F} = (P, Q)$ across this boundary is

$$\oint_{\partial D} \boldsymbol{F}\cdot\boldsymbol{n}\,\mathrm{d}s = \oint_{\partial D} P\,\mathrm{d}y - Q\,\mathrm{d}x.$$

Flux integrates the component normal to the boundary, whereas work integrates the tangential component. Apply Green to the field $(-Q, P)$, whose scalar curl is $\partial_x P + \partial_y Q$. We obtain the planar flux form

$$\oint_{\partial D} \boldsymbol{F}\cdot\boldsymbol{n}\,\mathrm{d}s = \iint_D \operatorname{div}\boldsymbol{F}\,\mathrm{d}A, \qquad \operatorname{div}\boldsymbol{F} = \partial_x P + \partial_y Q. \tag{41.1}$$

The same region and neighbourhood C^1 hypotheses as Green are required; its proved or sketch status is inherited without change.

For $\boldsymbol{F} = (2x, -y)$ on $D = \{x^2/4 + y^2 \le 1\}$, divergence is 1. The outward flux is therefore the ellipse area 2π [P]. To check the normal convention directly, use $x = 2\cos t$, $y = \sin t$. The flux integrand $Py' - Qx'$ is $4\cos^2 t - 2\sin^2 t$, whose full-turn integral is 2π. Near the top and bottom the field points inward, while near the two horizontal ends it points outward. Flux is their signed balance.

41.3 A regular surface patch and its area

Let E be an admissible compact planar parameter region and let $\boldsymbol{r}(u, v)$ be C^1 on a neighbourhood of E. Assume the map is injective on the parameter interior and

$$\boldsymbol{r}_u(u, v) \times \boldsymbol{r}_v(u, v) \neq \boldsymbol{0} \quad \text{at every interior point.}$$

Its image is a *regular surface patch*. Boundary seams or pole coordinates are allowed when they represent only boundary identifications, as with spherical coordinates. Pieces meet only along their boundaries when we add them to form a piecewise smooth surface.

A small parameter rectangle has tangent edge vectors $\boldsymbol{r}_u\Delta u$ and $\boldsymbol{r}_v\Delta v$. Their parallelogram has area $\|\boldsymbol{r}_u \times \boldsymbol{r}_v\|\,\Delta u\Delta v$, by the cross-product rule. This tangent area factor defines the scalar surface element and integral:

$$\mathrm{d}S = \|\boldsymbol{r}_u \times \boldsymbol{r}_v\|\,\mathrm{d}u\,\mathrm{d}v, \qquad \iint_S h\,\mathrm{d}S = \iint_E h(\boldsymbol{r}(u, v))\,\|\boldsymbol{r}_u \times \boldsymbol{r}_v\|\,\mathrm{d}u\,\mathrm{d}v.$$

Here h is continuous on the compact surface; $h = 1$ gives its area. The integral on the right is a proper planar integral with continuous integrand, so its existence follows from M. This definition through the tangent factor is not a claim that the surface is a three-dimensional solid with a volume density.

The definition is independent of an admissible regular change of surface parameters. To verify this, let $(u, v) = \psi(s, t)$ be a C^1 coordinate bijection satisfying M, and put $\boldsymbol{R} = \boldsymbol{r} \circ \psi$. The matrix chain rule gives

$$\boldsymbol{R}_s = \boldsymbol{r}_u u_s + \boldsymbol{r}_v v_s, \qquad \boldsymbol{R}_t = \boldsymbol{r}_u u_t + \boldsymbol{r}_v v_t.$$

Bilinearity and antisymmetry of the cross product give

$$\boldsymbol{R}_s \times \boldsymbol{R}_t = (u_s v_t - u_t v_s)(\boldsymbol{r}_u \times \boldsymbol{r}_v) = \det J_\psi\,(\boldsymbol{r}_u \times \boldsymbol{r}_v). \tag{41.2}$$

Taking norms supplies exactly the absolute Jacobian factor in the planar change-of-variables rule. Applying M therefore gives the same scalar surface integral. This proves coordinate independence within the stated class of patches and transitions. For finitely many patches with disjoint interiors, add their integrals; common boundary curves are not counted as extra surface area.

41.4 Orientation and vector flux

An *orientation* chooses a continuously varying side of each regular surface piece, with consistent choices where pieces meet. In a parametrisation one choice of unit normal is

$$\boldsymbol{n}_+ = \frac{\boldsymbol{r}_u \times \boldsymbol{r}_v}{\|\boldsymbol{r}_u \times \boldsymbol{r}_v\|}; \qquad \boldsymbol{n}_- = -\boldsymbol{n}_+.$$

For a surface that bounds a solid, the *outward* normal points away from that solid. An open surface, such as a disc, needs an explicitly specified orientation, for example upward.

For a continuous vector field defined near an oriented surface, its *flux* is $\iint_S \boldsymbol{F} \cdot \boldsymbol{n}\,\mathrm{d}S$. If the chosen normal agrees with $\boldsymbol{r}_u \times \boldsymbol{r}_v$, the formula is

$$\iint_S \boldsymbol{F} \cdot \boldsymbol{n}\,\mathrm{d}S = \iint_E \boldsymbol{F}(\boldsymbol{r}(u,v)) \cdot (\boldsymbol{r}_u \times \boldsymbol{r}_v)\,\mathrm{d}u\,\mathrm{d}v. \tag{41.3}$$

If it disagrees, insert a minus sign. The vector area element already includes the area factor: multiplying Eq. (41.3) by its norm again would count that factor twice. Reversing orientation reverses flux but leaves scalar area unchanged. Equation (41.2) shows why exchanging the parameter order reverses the induced normal. Under an orientation-preserving transition the flux stays the same by M; under a reversing transition the cross product must be negated to retain the originally specified normal.

A level description also determines a normal when its gradient is nonzero. Suppose a regular patch lies in $H(x,y,z) = c$, with $H \in C^1$ near the patch and $\nabla H \neq \boldsymbol{0}$ there. Differentiating $H(\boldsymbol{r}(u,v)) = c$ in each parameter gives

$$\nabla H(\boldsymbol{r}) \cdot \boldsymbol{r}_u = 0, \qquad \nabla H(\boldsymbol{r}) \cdot \boldsymbol{r}_v = 0.$$

The two tangents are independent by regularity, so their perpendicular line is the line of their nonzero cross product. Thus the unit normals are $\pm\nabla H/\|\nabla H\|$. The sign still has to be checked geometrically. This argument identifies the normal of a given regular patch; it does not establish that a level set has such a patch without the relevant existence hypotheses. For a sphere, $H = x^2 + y^2 + z^2$ has gradient $2(x,y,z)$, which points outwards at every sphere point and is nonzero when $R > 0$.

For a graph $z = g(x,y)$ with $g \in C^1$ near a compact planar region D, take $\boldsymbol{r}(u,v) = (u,v,g(u,v))$. Then

$$\boldsymbol{r}_u \times \boldsymbol{r}_v = (-g_u, -g_v, 1), \qquad \mathrm{d}S = \sqrt{1 + g_u^2 + g_v^2}\,\mathrm{d}u\,\mathrm{d}v.$$

The third component is positive, so this parameter order gives upward orientation everywhere. For $\boldsymbol{F} = (P,Q,H)$, upward flux becomes

$$\iint_D \bigl(-P(u,v,g)g_u - Q(u,v,g)g_v + H(u,v,g)\bigr)\,\mathrm{d}u\,\mathrm{d}v.$$

This expression can be simpler than computing a unit normal and scalar surface element separately.

Figure 41.2 shows both normal choices in a meridian section of the graph used next.

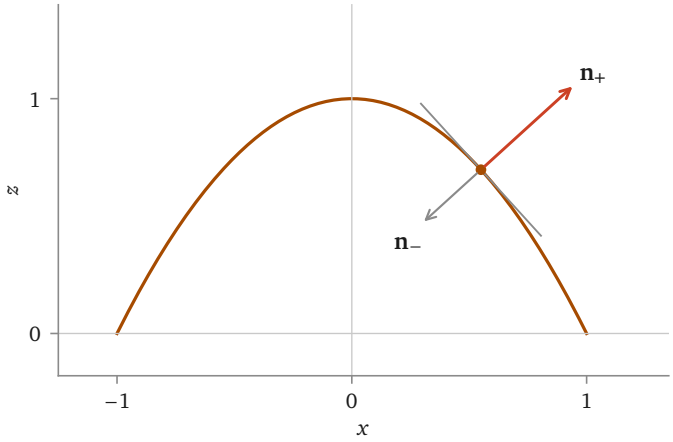


Figure 41.2: The section $y = 0$ of $z = 1 - x^2 - y^2$, with its tangent line and the two opposite normal directions. The upward choice has a positive vertical component.

Example 41.1 (A graph's area and upward flux)**.** Let S be the cap $z = 1 - x^2 - y^2$ over $x^2 + y^2 \le 1$, oriented upward, and let $\boldsymbol{F} = (0, 0, z)$. The graph is smooth, with vector area factor $(2x, 2y, 1)$. Its upward flux is

$$\iint_S \boldsymbol{F} \cdot \boldsymbol{n}\, dS = \iint_{x^2+y^2\le 1} (1 - x^2 - y^2)\, dA = 2\pi \int_0^1 (1 - r^2) r\, dr = \frac{\pi}{2}.$$

The scalar surface area uses a different integrand:

$$\operatorname{area}(S) = 2\pi \int_0^1 \sqrt{1 + 4r^2}\, r\, dr = \frac{\pi}{6}(5\sqrt{5} - 1).$$

For the last integral put $w = 1 + 4r^2$, so $dw = 8r\, dr$ and the bounds become 1 and 5. The downward flux is $-\pi/2$, while the area is unchanged.

41.5 A sphere with outward orientation

Parametrise the sphere of radius $R > 0$ using Lecture 39's angles:

$$\boldsymbol{r}(\theta, \varphi) = R(\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta), \quad 0 \le \theta \le \pi, \quad 0 \le \varphi \le 2\pi.$$

The radial unit vector is $\boldsymbol{e}_r = \boldsymbol{r}/R$. Direct differentiation and the cross product give

$$\boldsymbol{r}_\theta \times \boldsymbol{r}_\varphi = R^2 \sin\theta\, (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta) = R^2 \sin\theta\, \boldsymbol{e}_r.$$

It points outward for interior parameter values, where $\sin\theta > 0$. The poles and seam lie on the parameter boundary; the physical sphere itself is smooth there, as its graph patches in Lecture 39 show. Its area is

$$\int_0^{2\pi} \int_0^{\pi} R^2 \sin\theta\, d\theta\, d\varphi = 4\pi R^2.$$

For the field $\boldsymbol{F}(\boldsymbol{x}) = \kappa \boldsymbol{x}$, $\kappa \in \mathbb{R}$, the outward flux integrand is $\kappa R^3 \sin\theta$, giving

$$\iint_{\partial B_R} \boldsymbol{F} \cdot \boldsymbol{n}\, dS = 4\pi\kappa R^3.$$

Positive κ gives outward flow and negative κ inward flow. The result equals 3κ times the ball's volume. Lecture 42 will prove the relation behind that agreement for a box and state its broader compact-solid extension.

41.6 Divergence, curl and the Laplacian

For a C^1 field $\boldsymbol{F} = (F_1, F_2, F_3)$ on an open subset of $\mathbb{R}^3$, define

$$\operatorname{div} \boldsymbol{F} = \partial_x F_1 + \partial_y F_2 + \partial_z F_3,$$

$$\operatorname{curl} \boldsymbol{F} = (\partial_y F_3 - \partial_z F_2, \partial_z F_1 - \partial_x F_3, \partial_x F_2 - \partial_y F_1).$$

Divergence is scalar; curl is a vector. These are Cartesian coordinate formulas. Derivatives with respect to spherical variables cannot simply be substituted into them without transforming the formula. For a C^2 scalar function f, the Cartesian *Laplacian* is

$$\Delta f = \operatorname{div}(\nabla f) = \partial_x^2 f + \partial_y^2 f + \partial_z^2 f.$$

In the plane the corresponding definition uses the first two terms. A C^2 function with $\Delta f = 0$ is called *harmonic*. For example, $f = x^2 - y^2$ is harmonic in the plane since its two second partials are 2 and -2. For $\boldsymbol{F} = \kappa(x, y, z)$, divergence is 3κ and curl is zero, consistent with the flux computation just made.

Two identities from Schwarz

Two identities follow from Schwarz, with exactly its smoothness hypotheses. If $f \in C^2(U)$, the three components of $\operatorname{curl}(\nabla f)$ are differences of its mixed second partial derivatives, so

$$\operatorname{curl}(\nabla f) = \mathbf{0}.$$

If $\boldsymbol{F} \in C^2(U)$, expanding the other composition gives

$$\begin{aligned}\operatorname{div}(\operatorname{curl} \boldsymbol{F}) = {} & \partial_x \partial_y F_3 - \partial_x \partial_z F_2 + \partial_y \partial_z F_1 - \partial_y \partial_x F_3 \\ & + \partial_z \partial_x F_2 - \partial_z \partial_y F_1 = 0.\end{aligned}$$

Each pair involving the same component cancels by equality of its mixed partials. These are full pointwise proofs [P] on the open domain. They do not assert that every divergence-free field is the curl of a single C^1 field on the whole domain, or remove the domain condition in the scalar-potential theorem. The integral relations connecting divergence to outward flux and curl to boundary circulation are the subject of the next lecture.

41.7 Local source and circulation densities

The differential operators have a local integral interpretation that can be proved directly, before the general boundary laws. Let $\boldsymbol{F} \in C^1(U)$ and $a \in U$. For small $h > 0$ let Q_h be the cube centred at a with each coordinate increment between $-h$ and h. Its volume is $8h^3$, and its boundary is outward-oriented. We claim

$$\lim_{h \downarrow 0} \frac{1}{8h^3} \iint_{\partial Q_h} \boldsymbol{F} \cdot \boldsymbol{n} \, \mathrm{d}S = \operatorname{div} \boldsymbol{F}(a).$$

To prove this, pair the two faces perpendicular to the ith coordinate. For each fixed pair of tangential coordinates, their integrand difference is the integral of $\partial_i F_i$ along the normal segment, by the one-variable fundamental theorem. Given $\varepsilon > 0$, continuity makes $|\partial_i F_i(x) - \partial_i F_i(a)| < \varepsilon$ throughout the small cube, for all three i at once. The face difference thus differs from $2h\partial_i F_i(a)$ by at most $2h\varepsilon$. The tangential face area is $4h^2$, so its integrated error is at most $8h^3\varepsilon$. Adding the three pairs and dividing by cube volume bounds the total error by 3ε, proving the limit. This argument supplies a local density interpretation, using only coordinate segments and proper face integrals.

For curl, let C_h be the square of side $2h$ centred at a in the plane parallel to xy, with boundary anticlockwise as seen from positive z. The restricted field is C^1 on a neighbourhood of this square. Green and the planar line-integral definition give

$$\oint_{\partial C_h} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{x} = \iint_{C_h} (\partial_x F_2 - \partial_y F_1)\, \mathrm{d}A.$$

Subtract $(\operatorname{curl} \boldsymbol{F}(a))_3$ times its area $4h^2$. The difference in absolute value is at most $4h^2$ times the largest absolute variation of this continuous curl component on the square. After division by area that variation tends to zero. Consequently

$$\lim_{h \downarrow 0} \frac{1}{4h^2} \oint_{\partial C_h} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{x} = (\operatorname{curl} \boldsymbol{F}(a))_3.$$

The planes with ordered coordinates (y, z) and (z, x) give the first and second curl components in the same way, with positive normals along x and y, respectively. Their signs follow this cyclic coordinate order. Divergence therefore measures limiting outward flux per volume; the normal component of curl measures limiting circulation per area.

For example, the smooth field

$$\boldsymbol{F} = (-\omega y/2, \omega x/2, \lambda z), \qquad \omega, \lambda \in \mathbb{R},$$

has divergence λ and curl $(0, 0, \omega)$. Its source and circulation contributions can be varied independently. At the origin its cube flux is $8\lambda h^3$, and its positive horizontal square circulation is $4\omega h^2$, exactly realising the two limits. The interpretations concern these local integral limits, rather than assuming a global mechanical motion for an arbitrary field.

Summary 41.

- Green gives area $\oint x\,\mathrm{d}y = -\oint y\,\mathrm{d}x$ for positive boundary orientation under its stated hypotheses. Hence anticlockwise phase-plane traversal has $\oint p\,\mathrm{d}q = -\text{area}$.
- Planar outward flux is $\oint P\,\mathrm{d}y - Q\,\mathrm{d}x$ and equals the divergence area integral on Green's domains, with the same proof status.
- A regular surface patch has scalar area factor $\|\boldsymbol{r}_u \times \boldsymbol{r}_v\|$. Oriented flux uses the cross product itself, with its sign chosen to match the specified normal. Admissible reparametrisations preserve the resulting geometric integrals.
- Divergence is scalar, curl is vector and the Cartesian Laplacian is the sum of pure second partials. Schwarz proves $\operatorname{curl}\nabla f = \mathbf{0}$ for $f \in C^2$ and $\operatorname{div}\operatorname{curl}\boldsymbol{F} = 0$ for $\boldsymbol{F} \in C^2$.

Sheet 41 develops phase-plane area, planar and surface flux, and the polar and supplied spherical Laplacians. Lecture 42 compares surface integrals with the derivatives throughout a solid or along its boundary.

References

J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §§16.4–16.7.